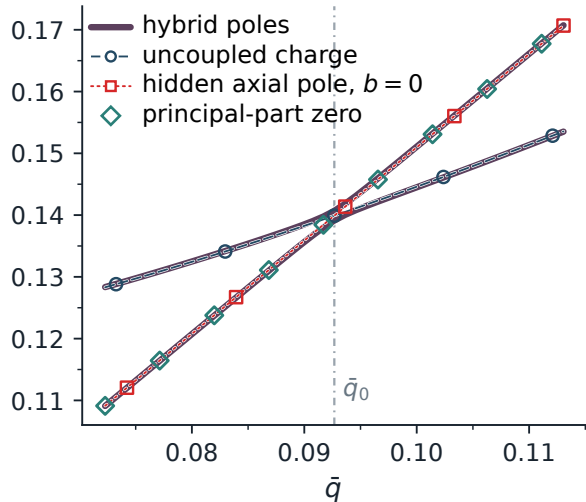


(a)

$$\max|\tilde{z}_0 - \bar{\omega}_a^0| = 1.2 \times 10^{-5}$$

(b)